

Complete-Case Analysis, Available-Case Analysis, and Weighting Methods for Missing Data

A Review with Proofs and Worked Examples

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Abstract

This report reviews classical methods for handling missing data when no model-based imputation is performed: complete-case (CC) analysis, available-case (AC) analysis, and weighted complete-case analysis, including propensity-score weighting and weighted generalized estimating equations (GEE). For each method we state the precision and bias properties, give worked numerical examples, and provide full derivations of the key theoretical results — in particular (i) the efficiency loss formula for complete-case means and regression coefficients under monotone missingness, (ii) the invariance of the complete-case odds ratio under additive (non-interactive) response mechanisms, (iii) the sufficiency of the propensity score for quasirandomization, and (iv) the bias-reducing role of weighting in generalized estimating equations. The treatment follows Chapter 3 of Little and Rubin, *Statistical Analysis with Missing Data*.

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1 Introduction and Motivation

When a dataset has missing values, the simplest strategies available — before resorting to model-based methods such as full-information maximum likelihood (FIML) or multiple imputation (MI) — are to either *discard* incomplete information or to *reweight* what remains so that it better represents the population. This report is organized around three such strategies:

- (1) **Complete-case (CC) analysis:** restrict the analysis to rows with no missing values at all.
- (2) **Weighted complete-case analysis:** keep only complete cases, but reweight them to correct for the fact that they may not be a random subsample of the full data.
- (3) **Available-case (AC) analysis:** for each variable (or pair of variables), use every case for which that variable (or pair) is observed, even if other variables on that row are missing.

None of these methods are generally recommended as a final solution; they are useful chiefly when the fraction of missing information is small, or as a baseline against which model-based methods (FIML, MI) are compared. The central theoretical question running through the chapter is: *under what conditions on the missingness mechanism does a given ad-hoc method remain unbiased, and at what cost in precision?*

2 Complete-Case Analysis

Complete-case (CC) analysis confines the entire analysis to the subset of rows with *no* missing values on any variable used in the analysis.

2.1 Advantages and Disadvantages

- **Advantage — simplicity:** any standard complete-data method (regression, ANOVA, etc.) applies without modification.
- **Advantage — comparability:** every statistic is computed on the same sample base, so cross-tabulated summaries are mutually consistent.
- **Disadvantage — loss of precision:** discarding rows reduces the effective sample size, increasing variance, *even if missingness is completely random.*

- **Disadvantage — bias:** if the complete cases are not a random subsample of all cases (i.e. the mechanism is not MCAR), CC estimates are biased for the population quantity of interest.

2.2 Quantifying the Loss of Precision

Let $\hat{\theta}_{CC}$ denote an estimate of a scalar parameter θ based on complete cases, and let $\hat{\theta}_{EFF}$ denote an efficient estimate based on *all* available data (e.g. a maximum-likelihood estimate that uses every observed value). Define the relative efficiency loss Δ_{CC} by

$$\text{Var}(\hat{\theta}_{CC}) = \text{Var}(\hat{\theta}_{EFF}) (1 + \Delta_{CC}). \quad (1)$$

Thus Δ_{CC} is the proportional increase in variance incurred by using CC analysis instead of the efficient method; $\Delta_{CC} = 1$ means CC analysis needs twice the variance of the efficient method, equivalent to halving the effective sample size.

Example Efficiency for Bivariate Normal Monotone Data

Consider bivariate normal data (Y_1, Y_2) with monotone missingness: Y_1 is observed for all n cases, while Y_2 is observed for only r of them ($n - r$ cases have Y_2 missing). Assume the data are MCAR, so that bias is not an issue and we are isolating the precision loss alone.

(a) Mean of Y_1 . CC analysis discards the $n - r$ cases with Y_2 missing even though their Y_1 values are perfectly good. Hence

$$\Delta_{CC}^* = \Delta_{CC} = \frac{n - r}{r}. \quad (2)$$

Derivation. The efficient estimator of $E[Y_1]$ uses all n observed values of Y_1 , so $\text{Var}(\hat{\theta}_{EFF}) = \sigma_1^2/n$. The CC estimator uses only the r complete cases, so $\text{Var}(\hat{\theta}_{CC}) = \sigma_1^2/r$. Then

$$1 + \Delta_{CC} = \frac{\text{Var}(\hat{\theta}_{CC})}{\text{Var}(\hat{\theta}_{EFF})} = \frac{\sigma_1^2/r}{\sigma_1^2/n} = \frac{n}{r} \implies \Delta_{CC} = \frac{n}{r} - 1 = \frac{n - r}{r}.$$

If half the cases are missing Y_2 (i.e. $r = n/2$), then $\Delta_{CC} = 1$: the variance of the CC mean of Y_1 is *doubled*, purely from discarding usable Y_1 values.

(b) Mean of Y_2 . Here CC analysis cannot avoid losing the $n - r$ cases with Y_2 itself missing, but it additionally fails to use the correlated information in their Y_1 values. With $\rho = \text{Corr}(Y_1, Y_2)$,

$$\Delta_{CC}^* = \frac{n - r}{r}, \quad \Delta_{CC} \approx \frac{(n - r)\rho^2}{n(1 - \rho^2) + r\rho^2}. \quad (3)$$

As $\rho^2 \rightarrow 0$, $\Delta_{CC} \rightarrow 0$ (no information lost, since Y_1 tells us nothing about Y_2). As $\rho^2 \rightarrow 1$, $\Delta_{CC} \rightarrow (n - r)/r = \Delta_{CC}^*$ (the incomplete cases become a perfect proxy for Y_2 , so failing to use Y_1 for them costs exactly as much as losing them outright).

(c) Regression coefficient of Y_2 on Y_1 . Here

$$\Delta_{CC}^* \approx \frac{n - r}{r}, \quad \Delta_{CC} = 0. \quad (4)$$

CC analysis is *fully efficient* for this estimand: cases with Y_1 observed but Y_2 missing contribute no information about the conditional distribution of Y_2 given Y_1 , since that conditional relationship can only be learned from cases where both are observed.

The efficiency loss from CC analysis is not a single fixed number for a dataset — it depends entirely on *which parameter* is being estimated. The same monotone-missing dataset can be wasteful for one estimand (the marginal mean of Y_1) and fully efficient for another (the regression slope of Y_2 on Y_1).

2.3 Quantifying the Bias

Example Bias of the Complete-Case Mean

Partition the population into respondents and non-respondents to Y , with population means μ_{CC} and μ_{IC} respectively, and let p_{CC} be the population proportion of respondents. The overall population mean is the mixture

$$\mu = p_{CC} \mu_{CC} + (1 - p_{CC}) \mu_{IC}.$$

The bias of the CC sample mean as an estimator of μ is

$$\begin{aligned} \mu_{CC} - \mu &= \mu_{CC} - [p_{CC} \mu_{CC} + (1 - p_{CC}) \mu_{IC}] \\ &= \mu_{CC}(1 - p_{CC}) - (1 - p_{CC}) \mu_{IC} \\ &= (1 - p_{CC})(\mu_{CC} - \mu_{IC}). \end{aligned} \tag{5}$$

That is, *bias* = (*fraction missing*) $\times$ (*difference between respondent and non-respondent means*). If the mechanism is MCAR, then $\mu_{CC} = \mu_{IC}$ and the bias is exactly zero.

Numerical illustration. Suppose 30% of respondents refuse to disclose income ($1 - p_{CC} = 0.3$), and refusers earn on average \$20,000 more than disclosers ($\mu_{CC} - \mu_{IC} = -20$, in thousands). Then the bias is $0.3 \times (-20) = -6$: the CC mean understates population income by \$6,000.

Example Bias of Complete-Case Regression Coefficients

Consider regressing Y on covariates $X_1, \dots, X_p$ where the regression function is correctly specified, with missingness possible in Y and/or the X 's. Let M denote the missing-data indicator ($M = 1$ if incomplete). The CC regression coefficients are **unbiased** provided

$$\Pr(M = 1 \mid X_1, \dots, X_p, Y) = \Pr(M = 1 \mid X_1, \dots, X_p), \tag{6}$$

i.e. missingness may depend on the covariates in any way, but must not depend further on Y once the covariates are conditioned upon (Glynn and Laird, 1986). Notably, this class of mechanisms includes some *non-ignorable* mechanisms, e.g. where the probability a covariate is missing depends on that covariate's own (unobserved) value — because the analysis conditions on the covariates anyway. CC regression coefficients *are* biased if missingness depends on Y even after conditioning on the covariates.

2.4 Bias of the Complete-Case Odds Ratio

Example Odds Ratios and Case-Control Validity

Let Y_1, Y_2 be dichotomous, and let M be the response indicator, with $M = 1$ if both are observed (complete case) and $M = 0$ otherwise. Suppose the log probability of being a complete case is **additive** in Y_1 and Y_2 :

$$\log \Pr(M = 1 \mid Y_1 = j, Y_2 = k) = \alpha + g_1(j) + g_2(k), \quad j, k \in \{0, 1\}, \quad (7)$$

with no interaction term between Y_1 and Y_2 . Then the complete-case odds ratio equals the population odds ratio exactly.

Proof. Exponentiating (7),

$$\Pr(M = 1 \mid Y_1 = j, Y_2 = k) = c a_j b_k, \quad c = e^\alpha, \quad a_j = e^{g_1(j)}, \quad b_k = e^{g_2(k)},$$

so the response probability factors into a row effect a_j (depending only on Y_1) and a column effect b_k (depending only on Y_2). Let $\pi_{jk} = \Pr(Y_1 = j, Y_2 = k)$ denote the true population cell probabilities. The probability of landing in cell (j, k) and being a complete case is

$$\Pr(Y_1 = j, Y_2 = k, M = 1) = \pi_{jk} \cdot c a_j b_k,$$

so the (rescaled) complete-case cell probabilities satisfy $\pi_{jk}^{CC} \propto \pi_{jk} a_j b_k$. The population odds ratio is $OR = \frac{\pi_{11}\pi_{00}}{\pi_{10}\pi_{01}}$. Computing the same ratio from the complete-case probabilities,

$$OR_{CC} = \frac{\pi_{11}^{CC} \pi_{00}^{CC}}{\pi_{10}^{CC} \pi_{01}^{CC}} = \frac{(\pi_{11} a_1 b_1)(\pi_{00} a_0 b_0)}{(\pi_{10} a_1 b_0)(\pi_{01} a_0 b_1)}.$$

Collect the nuisance factors in the numerator and denominator separately:

$$\text{numerator factor: } a_1 b_1 \cdot a_0 b_0 = a_0 a_1 b_0 b_1, \quad \text{denominator factor: } a_1 b_0 \cdot a_0 b_1 = a_0 a_1 b_0 b_1.$$

These are identical, so they cancel:

$$OR_{CC} = \frac{\pi_{11}\pi_{00}}{\pi_{10}\pi_{01}} = OR. \quad \blacksquare$$

Interpretation. The odds ratio is, by construction, invariant to independent rescaling of the rows and columns of a 2×2 table. An additive-on-the-log-scale response mechanism is exactly such a row/column rescaling, so the odds ratio is mathematically blind to it. This is the formal justification for **case-control sampling** in epidemiology: sampling diseased and healthy individuals at very different rates is precisely a row effect a_j depending only on disease status Y_1 (with $b_0 = b_1$, i.e. no dependence on exposure Y_2), which is a special case of additivity — so the odds ratio for exposure remains valid even though the case-control design grossly distorts the marginal distribution of disease status.

Where it fails. If there is a genuine *interaction* — e.g. individuals with $(Y_1, Y_2) = (1, 1)$ are disproportionately more or less likely to respond than the additive model predicts — an extra factor $d_{jk} = e^{g_{12}(j,k)}$ appears that does not cancel between numerator and denominator, and OR_{CC} is biased relative to OR .

2.5 Testing the MCAR Assumption

A discarded incomplete case still has some recorded variables, which can be used to assess MCAR: compare the distribution of a variable Y_j among complete cases against its distribution among incomplete cases for which Y_j is recorded. A significant difference indicates MCAR is implausible. Two limitations should be noted: such tests have low power when the incomplete-case sample is small, and they provide *no* direct evidence on whether the weaker MAR assumption holds, since MAR's defining condition involves the unobserved values themselves.

3 Weighted Complete-Case Analysis

If complete cases are suspected to be biased, they can be reweighted using information available for both respondents and non-respondents, so that the weighted sample better represents the full sampled population.

3.1 Two Layers of Probability

There are two distinct probabilistic stages between the population and the final analysis sample:

1. **Selection.** Unit i is included in the sample with probability π_i . This is determined by the sampling design and is **known**.
2. **Response.** Given selection, unit i responds with probability ϕ_i . This is a property of the unit's behaviour and is **unknown** — it must be estimated from data.

The joint probability of being selected *and* responding is $\pi_i\phi_i$, so each respondent represents $(\pi_i\phi_i)^{-1}$ population units.

3.2 The Horvitz–Thompson Estimator

If there were no nonresponse, a unit selected with probability π_i represents π_i^{-1} population units. The population total T is estimated by the Horvitz–Thompson estimator

$$t_{HT} = \sum_{i=1}^n y_i \pi_i^{-1}. \quad (8)$$

For stratified random sampling with n_j sampled from population stratum size N_j , this reduces to the stratified mean $\bar{y}_{st} = \frac{1}{N} \sum_j N_j \bar{y}_j$, with variance

$$\text{Var}(\bar{y}_{st}) = \frac{1}{N^2} \sum_j N_j^2 \left(\frac{1}{n_j} - \frac{1}{N_j} \right) S_{yj}^2.$$

3.3 Decomposing the Combined Weight

It is useful to see the weight $w_i \propto (\pi_i\phi_i)^{-1}$ as *two sequential corrections*, since selection and response solve conceptually different problems:

- **Step A (response correction, internal to the sample).** A respondent with response probability ϕ_i represents ϕ_i^{-1} *selected* units (themselves, plus the silent non-responders like them). This restores the composition of the originally selected sample.
- **Step B (selection correction, projecting to the population).** Each such reconstructed selected unit represents π_i^{-1} population units, exactly as in the no-nonresponse Horvitz–Thompson case.

Multiplying the two steps recovers the combined weight: $\phi_i^{-1} \times \pi_i^{-1} = (\pi_i \phi_i)^{-1}$. Crucially, the *population share* carried by a unit is determined entirely by π_i (Step B); the response correction ϕ_i^{-1} (Step A) only repairs sample composition *before* the population projection is applied — it does not, by itself, encode population share.

Numerical illustration. If $\pi_i = 0.1$ (1-in-10 selected) and $\phi_i = 0.5$ (half of similar selected units respond): Step A says this respondent stands in for $1/0.5 = 2$ selected units; Step B says each selected unit represents $1/0.1 = 10$ population units; total representation $= 2 \times 10 = 20$ population units.

3.4 Weighting Class Estimator

Divide the sample into J weighting classes using a variable C observed for both respondents and non-respondents. Let n_j be the sampled size and r_j the number of respondents in class j , with $r = \sum_j r_j$. The estimated response probability in class j is $\hat{\phi}_j = r_j/n_j$, giving a common weight $w_i = n_j/r_j$ to every respondent in class j . The weighting class mean is

$$\bar{y}_{wc} = n^{-1} \sum_{j=1}^J n_j \bar{y}_{jR}, \quad (9)$$

where $\bar{y}_{jR}$ is the respondent mean in class j and $n = \sum_j n_j$.

Where the weight is hidden in (9). Equation (9) contains no explicit symbol w_i , yet the weighting is present: if every respondent in class j receives weight $w_i = n_j/r_j$, then

$$\sum_{i \in j} w_i y_i = \frac{n_j}{r_j} \sum_{i \in j} y_i = \frac{n_j}{r_j} (r_j \bar{y}_{jR}) = n_j \bar{y}_{jR},$$

so the r_j in the weight cancels against the r_j respondents being summed, leaving exactly the term $n_j \bar{y}_{jR}$ appearing inside (9). The substitution of *sample share* n_j for *respondent share* r_j is the entire weighting adjustment, contrasted with the naive (unweighted) respondent mean $\bar{y}_{naive} = r^{-1} \sum_j r_j \bar{y}_{jR}$, which implicitly weights each class by how many people *responded* rather than how many were *sampled*.

Example Weighting Class Estimate of Mean Cholesterol

A health survey samples 100 individuals across four age strata:

Age group	n_j	r_j	Mean cholesterol	Response rate r_j/n_j
20–30	25	22	220	0.88
30–40	35	27	225	0.77
40–50	28	16	250	0.57
50–60	12	5	270	0.42

The naive respondent-only mean (averaging all 70 respondents directly) implicitly weights the 50–60 group by its respondent share $5/70 \approx 7\%$, even though this group is 12% of the sampled population — biasing the naive mean downward, since this group also has the highest cholesterol. The weighting class estimate corrects this:

$$\bar{y}_{wc} = \frac{1}{100} \left[25(220) + 35(225) + 28(250) + 12(270) \right] = \frac{5500 + 7875 + 7000 + 3240}{100} = 236.15.$$

Required assumption.

Assumption 1 (Quasirandomization). *Within weighting class j , respondents are a random sample of the sampled units — i.e. the data are MCAR within class j .*

Under Assumption 1, $\bar{y}_{wc}$ is unbiased, with variance (Oh and Scheuren, 1983)

$$\text{Var}(\bar{y}_{wc}) = \sum_j \left(\frac{n_j}{n} \right)^2 f_j S_j^2, \quad f_j = r_j^{-1} - N_j^{-1}.$$

3.5 Added Variance from Weighting

Weighting reduces bias but increases variance. For weights scaled to average one,

$$\text{Var}\left(\frac{1}{r} \sum_{i=1}^r w_i y_i\right) = \frac{\sigma^2}{r} [1 + \text{cv}^2(w_i)], \quad (10)$$

where $\text{cv}(w_i)$ is the coefficient of variation of the weights. The more unequal the weights, the larger the variance penalty: a few extreme-weight units can dominate and destabilize the estimate, analogous to a small number of loud voices overwhelming a survey.

3.6 Why Weighting Class Analysis Becomes Impractical with Many Covariates

Weighting classes can in principle be formed by crossing several fully-observed covariates simultaneously — there is no rule restricting the method to a single pivot variable. The practical limitation is combinatorial: crossing K categorical covariates multiplies the number of cells, and with realistic sample sizes this quickly produces cells with very few, or even *zero*, respondents, for which the weight n_j/r_j is undefined.

Example Sparsity with Many Covariates

Consider $n = 1000$ survey respondents with four fully-observed covariates: age group (3 levels), education (2), region (2), employment (2), giving $3 \times 2 \times 2 \times 2 = 24$ cells. Even though the average cell has roughly 42 people, realistic imbalance produces cells such as:

Cell	n_j	r_j	Weight n_j/r_j
Old, Degree, South, Unemployed	9	0	undefined (∞)
Old, No Degree, North, Unemployed	4	0	undefined (∞)
Middle, Degree, North, Unemployed	11	2	5.5 (unstable)

Cells with $r_j = 0$ make the weighting-class estimator undefined for those units; cells with very few respondents produce unstable, high-variance weights. This sparsity problem worsens rapidly as more covariates are crossed, and is unavoidable for continuous covariates, which cannot be crossed into cells without an arbitrary discretization step.

3.7 Propensity Score Weighting

Propensity weighting resolves the sparsity problem by replacing the multi-way cross-tabulation with a single estimated score.

3.7.1 Definition and Setup

Let X denote all variables observed for both respondents and non-respondents. Define the response propensity

$$p(x_i) = \Pr(m_i = 0 \mid x_i),$$

assumed strictly positive for all x_i (here $m_i = 0$ denotes response; $m_i = 1$ denotes missing — the labelling is a convention and the result is unaffected by which value is assigned to “complete”). Assume the data are **MAR**:

$$\Pr(M \mid X, Y) = \Pr(M \mid X). \quad (11)$$

3.7.2 The Sufficiency Theorem

Proposition 1 (Rosenbaum–Rubin propensity score sufficiency). *Under (11),*

$$\Pr(m_i = 0 \mid y_i, p(x_i)) = p(x_i) \quad \text{for all } x_i,$$

i.e. $\Pr[M \mid p(X), Y] = \Pr[M \mid p(X)]$: respondents form a random subsample of the sampled units within strata of the scalar propensity score $p(X)$, exactly as Assumption 1 requires with $C = p(X)$.

Proof. We proceed in four steps.

Step 1 (Tower rule). Since $p(x_i)$ is a deterministic function of x_i , it is coarser information than x_i itself; many distinct values of x_i may share the same $p(x_i)$. By the law of total expectation, conditioning on the finer variable x_i inside and averaging back out over $(y_i, p(x_i))$:

$$\Pr(m_i = 0 \mid y_i, p(x_i)) = E[\Pr(m_i = 0 \mid y_i, x_i) \mid y_i, p(x_i)].$$

This step is a pure probability identity and uses no assumption about the missingness mechanism.

Step 2 (Invoke MAR). By (11), $\Pr(m_i = 0 \mid y_i, x_i) = \Pr(m_i = 0 \mid x_i)$ — this is the *only* step in the proof where MAR is used:

$$E[\Pr(m_i = 0 \mid y_i, x_i) \mid y_i, p(x_i)] = E[\Pr(m_i = 0 \mid x_i) \mid y_i, p(x_i)].$$

Step 3 (Definition of $p(x_i)$). Substitute the definition $\Pr(m_i = 0 \mid x_i) = p(x_i)$:

$$E[\Pr(m_i = 0 \mid x_i) \mid y_i, p(x_i)] = E[p(x_i) \mid y_i, p(x_i)].$$

Step 4 (Conditioning on a known constant). Since $p(x_i)$ already appears in the conditioning set, $E[p(x_i) \mid y_i, p(x_i)] = p(x_i)$ (using the trivial identity $E[Z \mid Z] = Z$); the presence of y_i in the conditioning set is irrelevant once $p(x_i)$ is already known.

Chaining the four steps:

$$\Pr(m_i = 0 \mid y_i, p(x_i)) = p(x_i), \quad \text{for all } x_i. \quad \blacksquare$$

Interpretation. The proof shows that Y provides no additional information about response once $p(X)$ is known: $p(X)$ already encodes everything about X relevant to response (by its very definition), so conditioning further on the raw X cannot teach us anything about response beyond what $p(X)$ already captures, once MAR has stripped out Y ’s direct influence. This is structurally identical to the Rosenbaum–Rubin propensity-score theorem in causal inference, where conditioning on the propensity score balances covariates between treatment groups; here, conditioning on it makes respondents a random subsample.

Caveat. The theorem is proven for the *true*, population-level $p(X)$. In practice one uses an estimate $\hat{p}(X)$ from a fitted logistic regression, and the guarantee holds only to the extent that this logistic model is correctly specified.

3.7.3 Why Propensity Weighting Is Needed Despite Weighting Classes Already Existing

Weighting-class analysis is not restricted to a single covariate by any rule; it is restricted in practice by combinatorial sparsity (Example 3.6). Propensity weighting solves exactly this problem, for two reasons:

1. It allows *all* available covariates to contribute to the response model, without the curse of dimensionality of cross-tabulation, because a single logistic regression combines covariate effects additively rather than enumerating every joint cell.
2. It accommodates *continuous* covariates natively (e.g. income, years of education), which cannot be crossed into weighting-class cells without an arbitrary, unprincipled discretization step.

Example Propensity Weighting Resolving Sparsity

Returning to the $n = 1000$ savings-survey setting of Example 3.6, fit a single logistic regression

$$\text{logit } \hat{p}(x_i) = \hat{\alpha} + \hat{\beta}_1(\text{Age}_i) + \hat{\beta}_2(\text{Education}_i) + \hat{\beta}_3(\text{Region}_i) + \hat{\beta}_4(\text{Employment}_i),$$

using all 1000 sampled units. Even a unit whose exact covariate combination had zero respondents (e.g. “Old, Degree, South, Unemployed”) receives a well-defined predicted propensity (e.g. $\hat{p}(x_i) = 0.18$), borrowed from the additive pattern fitted across the entire sample rather than from its own (empty) cell. Sorting all units into five quintile buckets of $\hat{p}(x_i)$ yields well-populated groups:

Propensity bucket	n_j	r_j	Response rate
1 (lowest)	200	38	0.19
2	200	78	0.39
3	200	110	0.55
4	200	148	0.74
5 (highest)	200	174	0.87

No bucket has zero respondents, and the within-bucket weight $w_i = n_j/r_j$ (e.g. $200/38 = 5.26$ for bucket 1) is well-defined and stable for every unit. The weighting-class formula (9) is then applied directly to these buckets, exactly as if the buckets were a single observed categorical variable.

3.7.4 Two Ways to Use the Propensity Score

- **Bucketed (stratified):** group units into 5–6 strata of $\hat{p}(X)$; recompute $\hat{\phi}_j = r_j/n_j$ within each bucket; every unit in a bucket shares that bucket’s weight $w_i = n_j/r_j$. This is robust to mild misspecification of the logistic model, since the model is used only to *rank/group* units, not to read off an exact numeric weight.
- **Direct (continuous):** weight each respondent individually by $w_i = 1/\hat{p}(x_i)$ (Cassel, Särndal and Wretman, 1983), with no bucketing. This relies much more heavily on correct specification of the logistic model: a unit with a spuriously small $\hat{p}(x_i)$ receives an extreme, possibly destabilizing weight.

3.8 The Selection Weight Is Untouched by Propensity Modelling

The propensity score $p(X) = \Pr(\text{respond} \mid X)$ models the *response* stage only. The *selection* probability π_i is fixed by the sampling design, independent of the propensity model, and must still be multiplied in separately:

$$w_i = \frac{1}{\pi_i} \times \frac{n_j}{r_j} \quad (\text{bucketed}), \quad \text{or} \quad w_i = \frac{1}{\pi_i} \times \frac{1}{\hat{p}(x_i)} \quad (\text{direct}).$$

For example, under simple random sampling with $\pi_i = 0.05$ for everyone, a unit in propensity bucket 1 of Example 3.7.3 receives total weight $w_i = 20 \times 5.26 = 105.2$: a factor of 20 from selection (the unit represents 20 population units by design) and a factor of 5.26 from response (units like it respond roughly once every 5.26 times). Under a stratified design, π_i would vary across strata independently of the propensity bucket, since selection and response are governed by entirely separate mechanisms.

3.9 Post-Stratification and Raking

When the true population class proportions N_j/N are known from an external source (e.g. a census), the **post-stratified** mean

$$\bar{y}_{ps} = \frac{1}{N} \sum_j N_j \bar{y}_{jR} \quad (12)$$

replaces the sample-estimated proportion n_j/n with the known proportion N_j/N , generally achieving lower variance than the weighting-class estimator. When only the *marginal* totals of two cross-classifying factors are known externally (not their joint distribution), **raking** adjusts the observed cell counts n_{jl} by iterative proportional fitting — alternately scaling rows and columns until the row and column marginals match the known external totals, while preserving the association structure observed in the sample.

3.10 Summary of Weighting Methods

Weighting reduces bias from complete-case analysis by assigning the same correction weight to all variables measured for a given case. Because it discards incomplete cases entirely and provides no inherent variance control, weighting is most useful when covariate information is limited and the sample size is large, so that bias is the dominant concern relative to variance.

4 Available-Case Analysis

Available-case (AC) analysis uses, for each variable separately, every case for which that variable is observed — regardless of whether other variables on the same row are missing.

4.1 Motivation: The Cost of Requiring Full Completeness

Example Rapid Collapse of the Complete-Case Sample

If a dataset has 20 variables, each independently missing with probability 10%, the probability that a given row is fully complete is

$$0.9^{20} \approx 0.12.$$

Only about 12% of rows — representing roughly $12/0.9 \approx 13\%$ of all observed data values — survive complete-case analysis, even though each individual variable is only modestly affected by missingness.

This motivates available-case analysis, which uses every observed value of every variable, at the cost of a sample base that changes from variable to variable.

4.2 Pairwise Available-Case Covariances and Correlations

For a pair of variables (Y_j, Y_k) , let I_{jk} be the set of $n^{(jk)}$ cases with both observed. The pairwise covariance is

$$s_{jk}^{(jk)} = \sum_{i \in I_{jk}} (y_{ij} - \bar{y}_j^{(jk)})(y_{ik} - \bar{y}_k^{(jk)}) / (n^{(jk)} - 1). \quad (13)$$

Two competing ways to convert this into a correlation are:

$$r_{jk}^* = s_{jk}^{(jk)} / \sqrt{s_{jj}^{(j)} s_{kk}^{(k)}}, \quad (14)$$

$$r_{jk}^{(jk)} = s_{jk}^{(jk)} / \sqrt{s_{jj}^{(jk)} s_{kk}^{(jk)}}. \quad (15)$$

In (14), the variances $s_{jj}^{(j)}, s_{kk}^{(k)}$ are computed over each variable's *own* (possibly different) available-case sample base; this estimator can fall **outside** $[-1, 1]$, since numerator and denominator are computed on inconsistent sample bases. In (15) (Mathai, 1951), the variances are restricted to the *same* base I_{jk} as the covariance, which guarantees $r_{jk}^{(jk)} \in [-1, 1]$, by the same logic that an ordinary Pearson correlation on a single fixed sample cannot exceed unity in magnitude.

4.3 The Three-Variable Correlation Paradox

Example A Non-Positive-Definite Correlation Matrix

Consider the following artificial dataset of three variables across 12 observations (“?” denotes missing):

Y_1	1	2	3	4	1	2	3	4	?	?	?	?
Y_2	1	2	3	4	?	?	?	?	1	2	3	4
Y_3	?	?	?	?	1	2	3	4	4	3	2	1

Each pair of variables is jointly observed on a disjoint block of 4 rows. Applying the *range-safe* estimator (15) to each pair yields

$$r_{12} = 1, \quad r_{13} = 1, \quad r_{23} = -1.$$

This is logically impossible: if Y_1 and Y_2 move in perfect lockstep, and Y_1 and Y_3 also move in perfect lockstep, then Y_2 and Y_3 must move together as well (correlation +1), not in opposition. The contradiction arises because each pairwise correlation in (15) is computed from a *different* 4-row subset, with no row in which all three variables are jointly observed to anchor the three estimates to a single consistent dataset.

Even the range-safe pairwise correlation estimator (15), although individually confined to $[-1, 1]$, can produce a *correlation matrix* that is not positive semi-definite once three or more variables are involved. Since regression and most multivariate procedures require a positive-definite covariance matrix, this is not a cosmetic issue — it can make downstream analysis break entirely, requiring ad hoc repair of the estimated matrix.

4.4 Relative Efficiency of CC versus AC Analysis

Simulation evidence is mixed and does not favour one method universally:

- Kim and Curry (1977): AC analysis outperforms CC analysis when the data are MCAR and correlations between variables are modest.
- Haitovsky (1968); Azen and Van Guilder (1981): CC analysis outperforms AC analysis when correlations are large.

Neither method is generally satisfactory, which historically motivated the development of likelihood-based (FIML) and imputation-based (multiple imputation) methods that use all of the data through an explicit probability model, rather than ad hoc deletion or pairwise reweighting.

5 Weighted Generalized Estimating Equations

The generalized estimating equation (GEE) framework unifies the regression-based results of the preceding sections — complete-case bias conditions and weighting corrections — into a single general statement that does not require a fully specified likelihood.

5.1 Setup

Let $y_i = (y_{i1}, \dots, y_{iK})$ be a K -dimensional outcome vector for unit i , fully observed for $i = 1, \dots, r$ and missing or partially observed for $i = r + 1, \dots, n$. Let $x_i = (x_{i1}, \dots, x_{ip})^T$ be fully-observed covariates with mean model $E[y_i | x_i] = g(x_i, \beta)$ for unknown $\beta \in \mathbb{R}^d$, and let $z_i = (z_{i1}, \dots, z_{iq})^T$ be fully-observed *auxiliary* variables that may predict missingness but are not part of the mean model g .

5.2 The Complete-Data Estimating Equation

In the absence of missing data, β solves

$$\sum_{i=1}^n D_i(x_i, \beta) [y_i - g(x_i, \beta)] = 0, \quad (16)$$

where $D_i(x_i, \beta)$ is a $(d \times K)$ matrix of known weighting functions and $y_i - g(x_i, \beta)$ is the residual vector. This is the direct generalization of the normal equations $\sum_i X_i^T (y_i - X_i \beta) = 0$ from ordinary least squares to vector outcomes and possibly nonlinear mean functions. Liang and Zeger (1986) show that solving (16) yields a consistent estimator of β under mild regularity conditions, without requiring a fully specified likelihood for y_i .

5.3 Naive Complete-Case GEE

With missing data, the simplest approach restricts the sum to the r complete cases:

$$\sum_{i=1}^r D_i(x_i, \beta) [y_i - g(x_i, \beta)] = 0. \quad (17)$$

This yields a consistent estimate of β provided

$$\Pr(m_i = 1 | x_i, y_i, z_i) = \Pr(m_i = 1 | x_i), \quad (18)$$

i.e. missingness may depend on x_i (already in the mean model) in any way, but must not depend on y_i or on the auxiliary variable z_i once x_i is conditioned upon. This is the GEE analogue of the regression bias condition in Example 2.3; here z_i enters as a potential source of bias if dropout depends on it.

5.4 Weighted GEE

Weighted GEE (Robins, Rotnitzky and Zhao, 1995) introduces a weight $w_i(\hat{\alpha})$ multiplying each complete case's contribution:

$$\sum_{i=1}^r w_i(\hat{\alpha}) D_i(x_i, \beta) [y_i - g(x_i, \beta)] = 0, \quad (19)$$

where $w_i(\hat{\alpha}) = 1/p(x_i, z_i, \hat{\alpha})$ and $p(x_i, z_i, \hat{\alpha})$ is the estimated probability of being a complete case, obtained from a logistic regression of m_i on *both* x_i and z_i , with $\hat{\alpha}$ the maximum-likelihood logistic regression coefficients. This is precisely the propensity-score construction of Proposition 1, transplanted into the regression-equation setting, with the propensity model now explicitly including the auxiliary variable z_i .

If the logistic model is correctly specified, (19) yields a consistent estimate of β under the weaker condition

$$\Pr(m_i = 1 \mid x_i, y_i, z_i) = \Pr(m_i = 1 \mid x_i, z_i). \quad (20)$$

5.5 Comparison of Conditions

	Naive CC-GEE (18)	Weighted GEE (20)
Missingness may depend on	x only	x and z
Missingness must not depend on	y, z	y

Weighting absorbs the auxiliary variable z 's influence on dropout into the weight $w_i(\hat{\alpha})$, so the residual equation no longer “sees” that dependence — algebraically analogous to the cancellation of the row/column nuisance factors in the odds-ratio proof of Section 2.4. This relaxes the bias condition from “missingness depends only on x ” to “missingness may also depend on z , provided z is included in the propensity model.”

Example A Clinical Trial Dropout Scenario

Suppose a clinical trial studies the effect of drug dose (x_i) on blood pressure (y_i), and patients who report nausea in week 2 (z_i , recorded for everyone including eventual dropouts) are both more likely to drop out before the final measurement and more likely to have worse blood pressure outcomes. Naive CC-GEE, using only x_i , violates (18) (dropout depends on z_i) and is biased. Weighted GEE, modelling dropout as a function of both x_i and z_i , satisfies (20) and corrects the bias — even though nausea itself is never of scientific interest as a predictor of blood pressure.

5.6 Practical Caveats

- Both (20) and the propensity-score theorem rely on the logistic regression of m_i on (x_i, z_i) being correctly specified.
- Variance estimation must account for the fact that $\hat{\alpha}$ is itself estimated with sampling error; Robins, Rotnitzky and Zhao (1995) provide Taylor-series-based standard errors for this purpose.
- As emphasized throughout the chapter: *there is never any direct empirical evidence against MAR without assumptions*, so the plausibility of (18) or (20) must be argued from substantive knowledge of the data-collection process, not verified from the data alone.

6 Conclusion

Complete-case analysis is simple but can be arbitrarily wasteful of information — quantified by Δ_{CC} , which depends on the estimand, not just the dataset — and is biased whenever the probability of being a complete case depends on the outcome itself (after conditioning on observed covariates). Weighting corrects this bias by reweighting surviving complete cases according to an estimated response probability, at a variance cost governed by the dispersion of the weights; propensity scores extend this correction to settings with many, or continuous, auxiliary covariates, via a single dimension-reducing model whose sufficiency follows from a four-line application of the tower rule under MAR. Available-case analysis attempts to recover the information complete-case analysis discards, but introduces an internal-consistency problem: pairwise estimates computed from different sample sub-bases need not cohere into a valid joint covariance or correlation matrix. The generalized estimating equation framework unifies all of these regression-based bias and weighting results into a single equation, with the propensity-weighting construction appearing as a direct generalization of the survey-weighting ideas developed earlier in the chapter.

Collectively, these results motivate the likelihood-based and imputation-based methods — full-information maximum likelihood and multiple imputation — that use the full joint distribution of the data rather than ad hoc deletion or reweighting, and which form the subject of later chapters.

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